

- Q.24 Prepare event tree for flammable release.
 Q.25 Explain plant safety inspection.
 Q.26 Write about statutory system of accident reposting under the factories Act, 1948.
 Q.27 What are the objectives of safety survey and safety sampling.
 Q.28 Define Safety audit as per IS: 14489
 Q.29 Explain frequency severity indicator. How is it useful?
 Q.30 Define disabling injury.
 Q.31 What are the benefits of permit to work system?
 Q.32 Prepare event tree for busting of a pressure vessel.
 Q.33 Explain the importance of accident records.
 Q.34 What are the key facts of accident investigation.
 Q.35 Explain the guide words used in HAZOP study.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 How will you investigate the release of ammonia gas from ammonia plant of a fertilizer unit? Discuss.
 Q.37 Explain the various stages of BLEVE with a typical case example.
 Q.38 In a factory the average daily rate of employment is 900. There are 3 shifts of 8 hrs duration. In a year the factory works for 310 days. If the total reportable incident in the year was 60 and MDL was 120 days, find FR, SR, IR of accidents.

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Roll No.

1st Year / Industrial Safety Subject:- Appraisal Analysis Inspection and Control Procedures

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A lost time injury is _____?
 a) injury causing death
 b) serious bodily injury
 c) injury causing 24 hours or more disablement
 d) all of the above
 Q.2 Abbreviated form of _____ is F.R.
 a) Form Ratio b) Format Ratio
 c) Frequency Rate d) Frequency Ratio
 Q.3 Risk is defined as
 a) Probability of occurrence of the event X Severity of the event
 b) F.R. X S.R.
 c) Cannot be quantified
 d) None of the above
 Q.4 What is the first stage of risk assessment ?
 a) Exposure assessment
 b) Hazard identification
 c) Toxicity study
 d) Risk characterization

Q.5 Plant safety inspection is carried out with the help of

- a) Questionnaire b) Checklist
c) Either (a) or (b) d) None of the above

Q.6 Permit to work system is a _____

- a) Administrative Control
b) Engineering Control
c) Both (a) and (b)
d) None of the above

Q.7 The IS standard for guidelines for carrying out occupational health and safety audit is

- a) 1589 b) 14489
c) 3786 d) None of the above

Q.8 The IS standard available for computation of frequency and severity rates for industrial injuries is

- a) 2594
b) 3796
c) 3786
d) None of the given options

Q.9 Incidence rate is expressed as the number of injuries per _____ persons employed

- a) 1000
b) 2000
c) 5000
d) None of the given options

Q.10 The number of mandays lost charged for fatal injuries is _____

- a) 4500 b) 5000
c) 5500 d) 6000

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Event Tree Analysis helps in identifying the _____ of accidents.

Q.12 Frequency rate measures the frequency of accident occurrence per _____ man-hours worked.

Q.13 Safety sampling is carried out for observing the _____ operations.

Q.14 JSA is a systematic method _____.

Q.15 In plant safety inspection _____ observations are noted.

Q.16 HAZOP study is most beneficial if carried out during the _____ stage.

Q.17 Accident free period is the period between two consecutive reportable accidents. (True/False)

Q.18 The first element in the hierarchy of controls is _____

Q.19 Incidence rate is the ratio of number of accidents and average number of workers per _____ workers.

Q.20 Safety audit can be carried out by an auditor from outside the plant. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 What are the elements of permit to work system?

Q.22 What are the different measures to mitigate the effects of an emergency?

Q.23 Differentiate between FR and SR.